

Python Code

```
1  # Portfolio 1 : Expressions
2  # This is a code for applying coding expressions revolving about the stock market. The
   function of this code is to greet the user then get their username account, which
   includes
3  # their owned stocks and current net worth. The code asks them what service would they
   like to perform (buy stocks, sell stocks, check
4  # current stocks and their current value in the market, etc). in buy stocks functions,
   it shows them the stocks available and their
5  # individual price then ask them the amount of stocks they would like to purchase
   (this will then reduce their net worth for buying
6  # the stocks). for sell stocks, this asks them what stocks would they like to sell and
   how many and show the amount of money they would
7  # gain from selling (this will then increase their net worth after selling). In the
   owned stocks function, show their current value in
8  # the market then prediction on how the trend will go in the coming days
9
10 # this function allows my portfolio to have an unpredictable outcome, for which in
   this case is the trend of the stock market
11 import random
12
13 # Stock market data
14
15 # these are the available stocks in the market, and yes they are inspired from famous
   restaurants here in bicol
16 market = {
17     "JABE" : 180,
18     "MCDO" : 250,
19     "CHWK" : 140,
20     "BIGGS" : 130,
21     "INSL" : 320
22 }
23
24 # User Database
25
26 #this includes the basic information about the user
27
28 user = {
29     "username": "",
30     "cash": 10000, #starting money
31     "portfolio" : {} #owned stocks
32 }
33
34 # Helper Functions
35
36 # this includes the codes about calculations, stocks still available, or buy and sell
   of stocks
37 def showstocks(): #available stocks
38     print("\n Available Stocks:")
39     for stock, price in market.items():
40         print(f"{stock}: ₱{price}")
```

```

41
42 def showportfolio(): #money and stocks owned
43     print("\n Your Portfolio:")
44     totalvalue = user["cash"]
45
46     print(f"\n Cash: ₱{user['cash']}")
47     print (f"Net Worth: ₱{totalvalue}")
48
49     if not user["portfolio"]:
50         print("You don't currently have owned stocks.")
51     else:
52         for stock, qty in user["portfolio"].items():
53             currentprice = market[stock]
54             value = currentprice * qty
55             totalvalue += value
56             print(f"{stock}: {qty} shares | Current Value: ₱{value}")
57
58
59 # Market trend
60
61 # this includes the code for generating the trend of the stock market, though it is
62 # all just based on random result depending on the code
63
64 print("\n Market trend Prediction:")
65 for stock in market:
66     change = random.choice(['Uptrend, Downtrend, Stable'])
67     print(f"{stock}: {change}")
68
69 def buystocks():
70     showstocks()
71     stock = input("\nWhat stock do you want to buy? ").upper()
72
73     if stock not in market:
74         print("Invalid stock.")
75         return
76
77     try:
78         qty = int(input("How many shares do you want to procure? "))
79     except:
80         print("Invalid number.")
81         return
82
83     cost = market[stock] * qty
84
85     if cost > user["cash"]:
86
87         print("Not enough balance.")
88         return
89
90     user["cash"] -= cost
91     user["portfolio"][stock] = user["portfolio"].get(stock, 0) + qty
92
93     print(f"You bought {qty} shares of {stock} for ₱{cost} ")

```

```

94
95 def sellstocks():
96     if not user["portfolio"]:
97         print("You currently don't own any stocks.")
98         return
99
100     print("\nYour stocks:")
101     for stock, qty in user["portfolio"].items():
102         print(f"{stock}: {qty} shares")
103
104     stock = input("\nWhat stock(s) do you want to sell? ").upper()
105
106     if stock not in user["portfolio"]:
107         print("You don't have this stock.")
108         return
109
110     try:
111         qty = int(input("How many stock(s) do you want to sell? "))
112     except:
113         print("Invalid number.")
114         return
115
116     if qty > user["portfolio"][stock]:
117         print("You don't have enough stock(s)")
118         return
119
120     profit = market[stock] * qty
121     user["cash"] += profit
122     user["portfolio"][stock] -= qty
123
124     if user["portfolio"][stock] == 0:
125         del user["portfolio"][stock]
126
127     print(f"You sold {qty} share(s) of {stock} for ₦{profit}")
128
129 # Main program
130
131 #this will include the various inputs of the user interface. Including the username or
    account name, service would they like to perform (buy stocks, sell stocks, check
    current stocks and their current value in the market, etc)
132
133 print("Welcome to DaksStocks Sim!")
134
135 user["username"] = input("Enter your account name: ")
136 print(f"\nHello, {user['username']}! Starting balance: ₦{user['cash']}")
137
138 while True:
139     print("\n=====")
140     print("What would you like to do? ")
141     print("1 Buy Stocks")
142     print("2 Sell Stocks")
143     print("3 Check Portfolio")
144     print("4 Exit")
145     print("=====")

```

```
146
147     choice = input("Enter choice: ")
148
149     if choice == "1":
150         buystocks()
151     elif choice == "2":
152         sellstocks()
153     elif choice == "3":
154         showportfolio()
155     elif choice == "4":
156         print("Have a nice day! ")
157         break
158     else:
159         print("Invalid choice.")
```